National
Qualifications

X8477611

Mathematics
Paper 1 (Non-Calculator)

Marking Instructions

Please note that these marking instructions have not been standardised based on candidate responses. You may therefore need to agree within your centre how to consistently mark an item if a candidate response is not covered by the marking instructions.

Section 1

Question			Generic scheme	Illustrative scheme	Max mark
1.			•¹ use the discriminant	•¹ $3^2 - 4 \times k \times (-4)$	3
			•² apply condition and simplify	•² $9 + 16k = 0$ or $9 = -16k$	
			•³ determine the value of k	•³ $-\frac{9}{16}$	

Question			Generic scheme	Illustrative scheme	Max mark
2.			•1 start to differentiate •2 complete differentiation •3 evaluate	•1 $5\left(x^2+1\right)^4\ldots$ •2 $\ldots\times 2x$ •3 160	3

Question			Generic scheme	Illustrative scheme	Max mark
3.			**Method 1**	**Method 1**	3
			•1 equate composite function to x	•1 $f\left(f^{-1}(x)\right)=x$	
			•2 write $f\left(f^{-1}(x)\right)$ in terms of $f^{-1}(x)$	•2 $\dfrac{f^{-1}(x)+3}{2}=x$	
			•3 state inverse function	•3 $f^{-1}(x)=2x-3$	
			Method 2	**Method 2**	
			•1 write as $y=f(x)$ and start to rearrange	•1 $y=f(x)\Rightarrow x=f^{-1}(y)$ $2y=x+3$	
			•2 express x in terms of y	•2 $x=2y-3$	
			•3 state inverse function	•3 $f^{-1}(y)=2y-3$ $\Rightarrow f^{-1}(x)=2x-3$	

Question			Generic scheme	Illustrative scheme	Max mark
4.			•1 find gradient of first line •2 find gradient of second line •3 interpret results and state conclusion	•1 $-\frac{3}{2}$ •2 $\frac{2}{3}$ •3 $-\frac{3}{2}\times\frac{2}{3}=-1\Rightarrow$ lines are perpendicular.	3

Question			Generic scheme	Illustrative scheme	Max mark
5.	(a)	(i)	•1 calculate $\sin p$	•1 $\frac{3}{\sqrt{10}}$	**1**
		(ii)	•2 calculate adjacent side •3 calculate $\cos q$	•2 $\sqrt{6}$ •3 $\sqrt{\frac{3}{5}}$ or $\frac{\sqrt{3}}{\sqrt{5}}$	**2**
	(b)		•4 use addition formula •5 calculate remaining trig ratios and substitute into formula •6 calculate $\cos(p+q)$	•4 $\cos p \cos q - \sin p \sin q$ stated or implied by •5 •5 $\frac{1}{\sqrt{10}} \times \frac{\sqrt{6}}{\sqrt{10}} - \frac{3}{\sqrt{10}} \times \frac{2}{\sqrt{10}}$ •6 $\frac{\sqrt{6}-6}{10}$	**3**

Question			Generic scheme	Illustrative scheme	Max mark
6.	(a)		•1 interpret notation •2 state expression for $f\left(g\left(x\right)\right)$	•1 $f\left(x^2-2x\right)$ •2 $2\left(x^2-2x\right)+5$	**2**
	(b)		•3 state expression for $g\left(f\left(x\right)\right)$	•3 $\left(2x+5\right)^2-2\left(2x+5\right)$	**1**
	(c)		•4 express $g\left(f\left(x\right)\right)-f\left(g\left(x\right)\right)$ in standard quadratic form •5 identify common factor •6 complete the square •7 process for c and write in required form	•4 $2x^2+20x+10$ •5 $2\left(x^2+10x\ldots\right.$ stated or implied by •6 •6 $2\left(x+5\right)^2\ldots$ •7 $2\left(x+5\right)^2-40$	**4**

Question			Generic scheme	Illustrative scheme	Max mark
7.			•1 start to integrate •2 complete integration	•1 $6\sin\left(3x+\frac{\pi}{4}\right)\ldots$ •2 $\ldots\times\frac{1}{3}+c$	2

Question			Generic scheme	Illustrative scheme	Max mark
8.			•¹ use $m = \tan\theta$ •² evaluate exact value •³ determine equation	•¹ $m = \tan\frac{2\pi}{3}$ stated or implied by •² •² $-\sqrt{3}$ •³ $y = -\sqrt{3}x + 4\sqrt{3}$	3

Question			Generic scheme	Illustrative scheme	Max mark
9.	(a)		•¹ set $y = y$ and arrange in standard form •² factorise and state x-coordinate of A	•¹ $6x^2 - 18x = 0$ •² $x = 3$	**2**
	(b)		•³ know to integrate and interpret limits •⁴ use 'upper − lower' •⁵ integrate •⁶ substitute limits •⁷ evaluate	•³ $\int_0^3 \dots \, dx$ •⁴ $\int_0^3 \left(\left(x^3 - 7x^2 + 12x + 3\right) - \left(x^3 - x^2 - 6x + 3\right)\right) dx$ •⁵ $-\frac{6}{3}x^3 + \frac{18}{2}x^2$ •⁶ $\left(-2 \times 3^3 + 9 \times 3^2\right) - 0$ •⁷ 27 (units²)	**5**

Question			Generic scheme	Illustrative scheme	Max mark
10.			•[1] know to use (synthetic) division or substitution	•[1] eg $\begin{array}{c\|cccc} \ldots & 6 & -13 & 0 & 4 \\ \hline & 6 & & & \end{array}$ **OR** $6\times(\ldots)^3-13\times(\ldots)^2+4$	4
			•[2] complete process using a root	•[2] eg $\begin{array}{c\|cccc} 2 & 6 & -13 & 0 & 4 \\ & & 12 & -2 & -4 \\ \hline & 6 & -1 & -2 & 0 \end{array}$ **OR** $6\times(2)^3-13\times(2)^2+4=0$	
			•[3] identify quadratic factor	•[3] $6x^2-x-2$	
			•[4] express in fully factorised form	•[4] $(x-2)(3x-2)(2x+1)$	

Question			Generic scheme	Illustrative scheme	Max mark
11.	(a)		$\bullet^1$ state maximum value	$\bullet^1$ 7	**1**
	(b)	(i)	$\bullet^2$ state maximum value	$\bullet^2$ 13	**1**
		(ii)	$\bullet^3$ state value of x	$\bullet^3$ 10	**1**

Section 2

Part A

Question			Generic scheme	Illustrative scheme	Max mark
12.	(a)	(i)	•[1] identify components of $\overrightarrow{AB}$ •[2] find $\left\lvert\overrightarrow{AB}\right\rvert$	•[1] eg $\begin{pmatrix} 3 \\ -6 \\ 6 \end{pmatrix}$ or $\sqrt{3^2+(-6)^2+6^2}$ •[2] 9	**2**
		(ii)	•[3] state ratio	•[3] $3:2$	**1**
	(b)		•[4] determine coordinates	•[4] $(9,-8,5)$	**1**

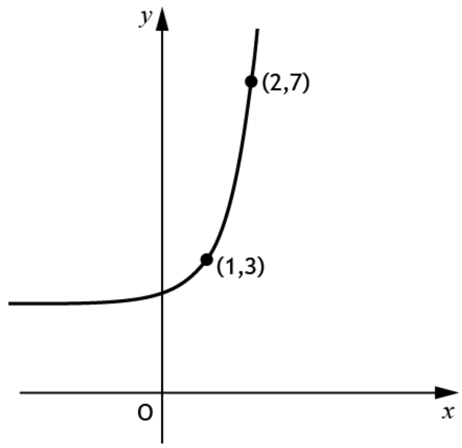
Question			Generic scheme	Illustrative scheme	Max mark
13.	(a)		•1 begin to find u_6 •2 determine u_5	•1 $20 = \frac{2}{3}u_6 + 8$ •2 $u_5 = 15$	**2**
	(b)		•3 know how to find limit •4 evaluate limit	•3 $L = \frac{2}{3}L + 8$ or $L = \frac{8}{1-\frac{2}{3}}$ •4 24	**2**

Question			Generic scheme	Illustrative scheme	Max mark
14.			•¹ expand •² calculate $\mathbf{u.u}$ or $\mathbf{u.v}$ •³ calculate $\mathbf{u.v}$ or $\mathbf{u.u}$ and complete calculation	•¹ $\mathbf{u.u}+\mathbf{u.v}$ •² 16 or -10 •³ -10 or 16 leading to 6	3

Part B

Question			Generic scheme	Illustrative scheme	Max mark
15.			•[1] determine radius of circle •[2] find coordinates of P •[3] state equation of circle	•[1] 2 stated or implied by •[3] •[2] $(8,7)$ stated or implied by •[3] •[3] $(x-8)^2+(y-7)^2=4$	**3**

Question			Generic scheme	Illustrative scheme	Max mark
16.			•¹ use laws of logarithms •² use laws of logarithms •³ use laws of logarithms •⁴ evaluate expression	•¹ $\log_2 3^2$ stated or implied by •³ •² $\log_2 (12\times6)$ stated or implied by •³ •³ $\log_2 \frac{12\times6}{3^2}$ •⁴ 3	**4**

Question			Generic scheme	Illustrative scheme	Max mark
17.	(a)		•¹ in first quadrant, graph reflected in $y = x$ passing through $(1,3)$ and $(2,7)$ •² in second quadrant, graph approaching $y = 2$ from above	•¹, •² 	**2**
	(b)		**Method 1** •³ interpret information and start to solve •⁴ solve and state coordinates **Method 2** •³ determine $f^{-1}(x)$ •⁴ evaluate $f^{-1}(0)$ and state coordinates	**Method 1** •³ $\log_5(x-2)+1=0$ leading to $\log_5(x-2)=-1$ •⁴ $\left(0,\frac{11}{5}\right)$ **Method 2** •³ $5^{(x-1)}+2$ •⁴ $\left(0,\frac{11}{5}\right)$	**2**

[END OF MARKING INSTRUCTIONS]